1 Social Optimum: Sioux Falls Network

1.1 Network and demand

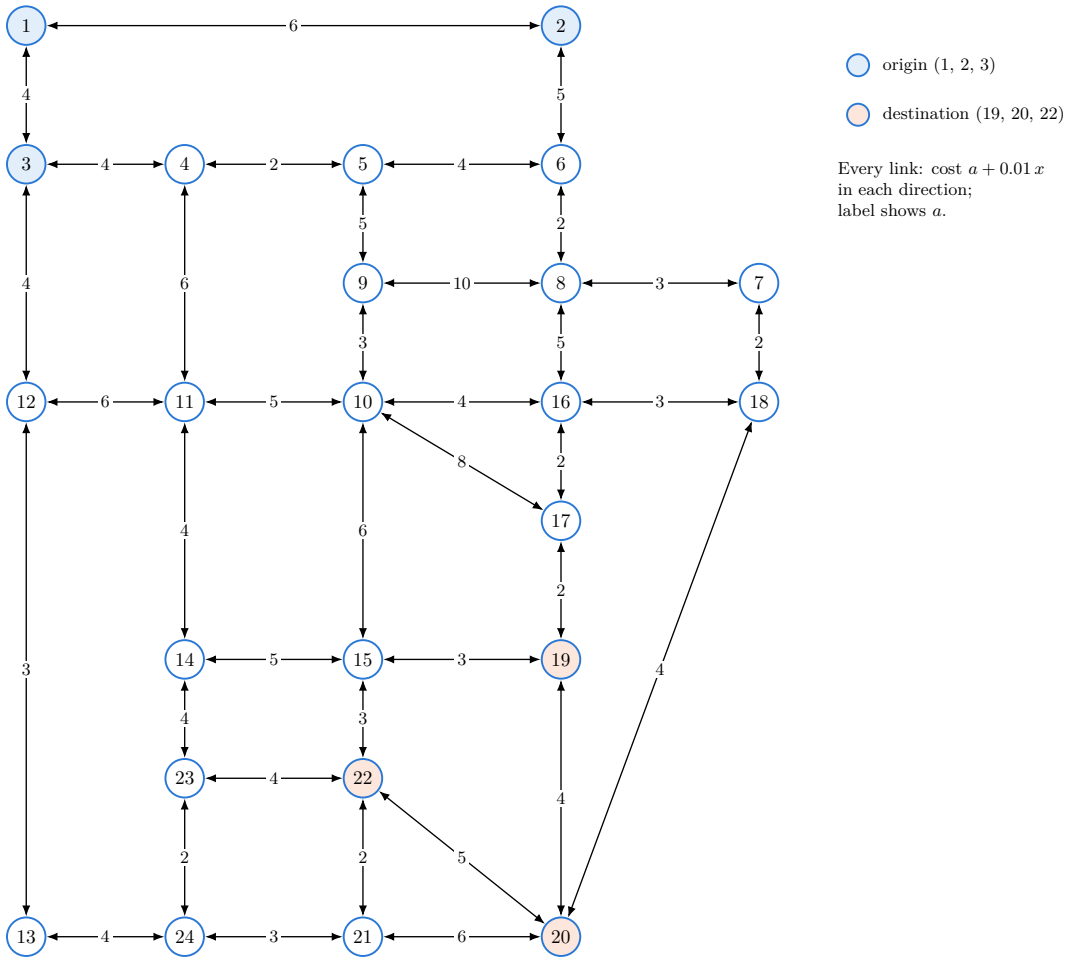


Figure 1: Sioux Falls network. Every link costs $a + 0.01x$ in each direction; labels show a.

There are 24 nodes and 76 directed edges. Each agent draws an origin from $\{1, 2, 3\}$ and a destination from $\{19, 20, 22\}$ using the run's seed, so demand varies by seed. With `forward_only` there are 404 feasible paths. We use $N = 250$, 500 and 1000.

1.2 Derivation

Timing matters. Edge $2 \to 6$ is the first edge for agents from 2 but the second for agents from 1 via 2, so its users can cross it in different steps. With $X_{e,t}$ agents crossing e in step t,

$$SC = \sum_{e,\,t} X_{e,t}\,(a_e + 0.01\,X_{e,t}),$$

minimised over how many agents of each origin–destination pair take each path. There is no closed form, so we bracket the optimum numerically:

- **Lower bound:** allow fractional agents and solve the resulting convex problem.

- **Upper bound:** round to whole agents, then move single agents between paths while that lowers the cost.

In all 153 cases (3 sizes × 51 seeds) the bounds are within 0.13 of each other, so we report the upper bound as SC_{opt}.

Data. Below, `<s>` is `sioux_falls_N250`, `sioux_falls_N500` or `sioux_falls`.

- Network and agent draws: `scenarios/<s>`, with the config seed (42) and the sweep seeds.
- Sweep seeds (3821–3870) and simulated costs (`social_cost`): `results/<s>/raw_results.csv`.
- Best α: lowest `mean_social_cost` in `results/<s>/summary.csv`.
- Static optimum: agrees with `src/benchmarks/frank_wolfe.py` (`--demand sampled`).

1.3 Results

	Seed 42				Sweep seeds
N	Lower bound	SC_{opt}	SC_{opt}/N	Congestion share	SC_{opt}/N (mean ± sd)
250	5400.25	5400.33	21.601	9.6%	21.773 ± 0.139
500	11714.81	11714.87	23.430	14.6%	23.572 ± 0.086
1000	26684.81	26684.91	26.685	23.4%	26.739 ± 0.062

Table 1: Social optimum. Congestion share is the part of the cost from congestion, $\sum 0.01\, X_{e,t}^2$.

	Simulator (mean of 50 trials)			
N	$\alpha = 0$	Best α	$\alpha = 1$	Static optimum
250	1.008	1.006 ($\alpha = 0.02$)	1.011	1.084
500	1.039	1.025 ($\alpha = 0.02$)	1.038	1.140
1000	1.108	1.055 ($\alpha = 0.01$)	1.161	1.222

Table 2: Cost relative to SC_{opt}. Each simulated trial is divided by the optimum for its own seed. The static optimum is for seed 42.

- The optimum *staggers* agents in time. At $N = 1000$, edge $2 \to 6$ carries 271, 139 and 17 agents at steps 0, 1 and 2, so later arrivals add nothing to earlier congestion. The static model cannot do this, so its optimum is 8–22% higher and unreachable in the simulator.
- The gap between the simulator and the optimum grows with N: at the best α it is 0.6%, 2.5% and 5.5%.
- Reference lines for the sweep graphs: 21.773, 23.572 and 26.739 per agent.